

Day 9 - Cleaning & Split Review

Data Quality Findings :

- Total samples: 399
- Total features: 6
- Missing values: 0
- Duplicate records: 0
- Invalid bit labels: 0
- Dataset is clean with no major quality issues detected.



Class Balance Findings :

Class	Count	Percentage
0	242	60.65%
1	157	39.35%

- Dataset shows a moderate class imbalance.
- Both classes are sufficiently represented for model training.

Proposed Split Strategy :

Dataset Split	Percentage
Training Set	70%
Validation Set	15%
Test Set	15%

Proposed Scaling Strategy :

- Drop S No column.
- Encode NTU if required.
- Apply StandardScaler for:
 - Analog Value
 - Analog Voltage
- Required for Logistic Regression, SVM, and MLP.
- Random Forest can be trained with or without scaling.